



NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations – December 2025

21EC504/20EC504 – VLSI DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Design the schematic circuit, stick diagram and layout for OAI31 and also estimate the area of the layout in terms of λ .	8	L*3	2	1
	b) Derive the Scaling effect on Oxide capacitance per unit area, Drain current and Power Dissipation in case of constant voltage scaling and constant field scaling.	6	L2	2	1
	c) Explain with the neat diagram VLSI Design flow in 3 Domains.	6	L2	1	1
2.	a) i) Implement the 1 bit Half adder using TGs also write truth table. ii) Implement Tristate buffer and Tristate Inverter. Also write truth table and symbol.	8	L3	1	1
	b) Explain the following terms i. Regularity ii. Modularity	6	L2	1	1
	c) Draw CMOS circuit for out = not(A(B+C)+DE) , with the help of Euler path draw stick diagram	6	L3	1,2	1
3.	a) With neat sketches of cross sectional view and associated mask, explain the fabrication of CMOS inverter in an n-well process.	10	L2	2	1
	b) Explain semi-custom and full-custom design along with advantages and disadvantages of each.	6	L2	1	1
	c) With neat sketch explain lamda based rules	4	L2	2	1
Unit – II					
4.	a) Write any 6 circuit diagrams which illustrate the improved latches and state the feature of the each circuit.	9	L2	4	1
	b) Define the following terms i. Rise time ii. Fall time iii. Propagation delay iv. Contamination delay	6	L2	3	1
	c) Derive the expression for the minimum delay for N stage logic cascade with branches, using logical effort technique.	5	L2	3	1

P.T.O.

5. a) For the 6 – input AND implementation shown in figure 5(a) Develop an expression for the delay of each path if the path electrical effort is H. What design is fastest for $H = 1$? For $H = 5$? Explain your conclusions intuitively.

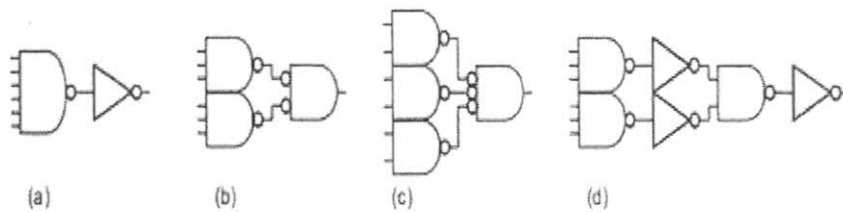


Figure 5(a): 6 input AND gates

- | | | | | |
|--|---|----|---|---|
| | 8 | L3 | 3 | 1 |
|--|---|----|---|---|
- b) Derive the expression for threshold voltage of a static CMOS inverter. Mention the different regions of operations for nMOS and pMOS.
- | | | | | |
|--|---|----|---|---|
| | 6 | L2 | 3 | 1 |
|--|---|----|---|---|
- c) Draw neat circuit diagram along with the truth table for the following circuit
- i) OAI321 Pseudo nMOS circuit.
- ii) 2 input EXOR gate in CVSL
- | | | | | |
|--|---|----|---|---|
| | 6 | L3 | 4 | 1 |
|--|---|----|---|---|
6. a) Design CMOS positive edge triggered D flip-flop and explain its working.
- | | | | | |
|--|---|----|---|---|
| | 8 | L2 | 4 | 1 |
|--|---|----|---|---|
- b) Write the schematic of 3 input NAND gate having charging path equal to discharge path resistance and its RC equivalent circuit. Estimate the t_{pdr} , t_{pdf} of it, if the output is loaded with "h" identical NAND3 gates.
- | | | | | |
|--|---|----|---|---|
| | 6 | L3 | 3 | 1 |
|--|---|----|---|---|
- c) Consider the two designs for two input OR gate as shown in figure Q.6(c). Comment which design is better and justify your answer by computing the path effort, minimum delay.

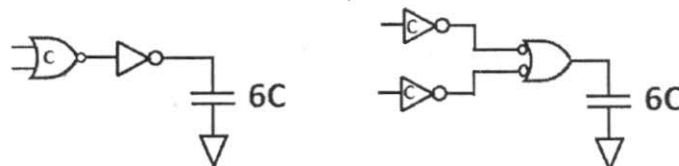


Figure Q6(c): 2 input OR gate

6	L3	3	1
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Unit – III

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|--|----|----|---|---|
| 7. a) Analyze the working of 3 Transistor DRAM cell with pull up and read/ write circuits. | 10 | L2 | 5 | 1 |
| b) Explain read and write operation in a 3T DRAM cell. | 10 | L3 | 5 | 1 |
| 8. a) Explain any 5 manufacturing test principles. | 10 | L2 | 5 | 1 |
| b) Explain with neat block diagram scan based approach of design for testability With neat diagram explain BIST. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
